MTH107 2022S2 Exam

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* MTH107 Calculus I
* MTH108 Calculus II
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* MTH319 Mathematical Methods I
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* MTH220 Statistical Methods and Inference
* MTH355 Basic Mathematical Optimisation

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MTH107

Timed Online Assignment - July Semester 2022

Calculus I

Tuesday, 15 November 2022 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and Not Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

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7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

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Answer all questions. (Total 100 marks)

Question 1a

Determine

$$\lim_{x \to 1} \frac{x^4 - 1}{x - 1}$$

<u>without using L'Hospital's Rule</u>. Show your steps clearly.

(6 marks)

Question 1b

Determine

$$\lim_{x \to 0} \frac{\sqrt{x+5} - \sqrt{5}}{x}$$

<u>without using L'Hospital's Rule</u>. Show your steps clearly.

(7 marks)

Question 1c

Show that there exists a positive real root for the equation $\sin x = x^3 - 2.$ Justify your answer.

(7 marks)

Question 2a

Let f be the function defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} \sin x \sin\left(\frac{1}{x^2}\right) & \text{if } x \neq 0; \\ 0 & \text{if } x = 0. \end{cases}$$

Discuss whether f is continuous at 0.

(7 marks)

Question 2b

Let f be the function defined by

$$f(x) = \begin{cases} \frac{ax^2-1}{x-1} & \text{if } x \neq 1; \\ b & \text{if } x = 1. \end{cases}$$

Find all possible values of $a, b \in \mathbb{R}$ such that f is continuous at $x = 1$. Justify your answer.

(7 marks)

Question 2c

Let $f(x) = |2x^3|$. Is f differentiable at 0? If yes, determine the values of $f'(0)$. Justify your answer.

(6 marks)

Question 3a

Determine

$$\lim_{x \to 0} \frac{\tan x - x}{x}.$$

Show your steps clearly.

(8 marks)

Question 3b

Show that

$$|e^{2x} - e^{2y}| \leq 8|x - y|$$

for all $x < y < \ln 2$.

(12 marks)

Question 4a

Let $f(x) = e^{x^3}$. Determine $f^{(60)}(0)$, expressing your answer in the form $\dfrac{a!}{b!}$.

(12 marks)

Question 4b

Determine the Taylor expansion of

$$\frac{1}{(1+x)^2}$$

about $x = 0$, up to and including the term in x^3. Use this Taylor expansion to estimate the value of $\frac{1}{1.01^2}$, giving your answer to 4 decimal places.

(8 marks)

Question 5

Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be the function defined by

$$f(x) = \frac{2 + x^2}{x^2 + 2x + 3}.$$

Question 5a

Apply the first derivative test to compute the intervals on which f is increasing.

(6 marks)

Question 5b

Determine the horizontal and vertical asymptotes of f if they exist. Justify your answers.

(7 marks)

Question 5c

Determine the critical point(s) of f. Apply relevant derivative tests and discuss the nature of these critical point(s). Show your steps clearly.

(7 marks)

----- END OF PAPER -----

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1(a) $\lim_{x \to 1} \frac{x^4-1}{x-1} = \lim_{x \to 1} \frac{(x^2-1)(x^2+1)}{x-1}$

$= \lim_{x \to 1} \frac{(x-1)(x+1)(x^2+1)}{x-1}$

$= \lim_{x \to 1} (x+1)(x^2+1)$

$= 4$

(b) $\lim_{x \to 0} \frac{\sqrt{x+5}-\sqrt{5}}{x} = \lim_{x \to 0} \frac{\sqrt{x+5}-\sqrt{5}}{x} \times \frac{\sqrt{x+5}+\sqrt{5}}{\sqrt{x+5}+\sqrt{5}}$

$= \lim_{x \to 0} \frac{(x+5)-5}{x(\sqrt{x+5}+\sqrt{5})}$

$= \lim_{x \to 0} \frac{x}{x(\sqrt{x+5}+\sqrt{5})}$

$= \lim_{x \to 0} \frac{1}{\sqrt{x+5}+\sqrt{5}}$

$= \frac{1}{2\sqrt{5}}$

(c) $\sin x = x^3 - 2$

$\sin x - x^3 + 2 = 0$

let $f(x) = \sin x - x^3 + 2$

$f(0) = \sin 0 + 0 + 2 = 2$

$f(\pi) = \sin \pi - \pi^3 + 2 = 2 - \pi^3$

By Intermediate Value Theorem,

since $f(0) > 0$ and $f(\pi) < 0$, $\exists c$ on $(0, \pi)$ s.t. $f(c) = 0$

There exists a positive real root.

2(a) $-|\sin x| \le -|\sin x|\,|\sin\frac{1}{x}| \le \sin x \sin\frac{1}{x} \le |\sin x|\,|\sin\frac{1}{x}| \le |\sin x|$

$$-|\sin x| \le \sin x \sin\frac{1}{x} \le |\sin x|$$

$$\lim_{x\to 0} -|\sin x| \le \lim_{x\to 0} \sin x \sin\frac{1}{x} \le \lim_{x\to 0} |\sin x|$$

$$0 \le \lim_{x\to 0} \sin x \sin\frac{1}{x} \le 0$$

By Squeeze Theorem, $\lim_{x\to 0} \sin x \sin\frac{1}{x} = 0$

$\therefore \lim_{x\to 0} f(x) = 0$

Since $\lim_{x\to 0} f(x) = f(0) = 0$, then $f(x)$ is continuous at $x=0$

(b) let $ax^2-1 = (x-1)(cx+d)$

$ax^2-1 = cx^2 + (d-c)x - d$

$\Leftrightarrow a=c,\ d-c=0,\ d=1$

$c=d=a=1$

$$\lim_{x\to 0} \frac{ax^2-1}{x-1} = \lim_{x\to 0} \frac{(x-1)(cx+d)}{x-1}$$

$$= \lim_{x\to 0} cx+d$$

$$= d$$

$$= 1$$

Since $f(x)$ is continuous, then $f(0) = \lim_{x\to 0} f(x)$

$\Rightarrow b = d = 1$

$\therefore a = 1$

$\therefore b = 1$

2(c) $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} |2x^3| = \lim_{x \to 0^+} 2x^3 = 0$

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} |2x^3| = \lim_{x \to 0^-} -2x^3 = 0$

$f(0) = 0$

$\therefore$ $f(x)$ is continuous at $x = 0$

$$\lim_{h \to 0^+} \frac{f(0+h) - f(0)}{h - 0} = \frac{2h^3 - 0}{h} = 2h^2 = 0$$

$$\lim_{h \to 0^-} \frac{f(0+h) - f(0)}{h - 0} = \frac{-2h^3 - 0}{h} = -2h^2 = 0$$

$$\therefore \lim_{h \to 0} \frac{f(h) - f(0)}{h - 0} = 0$$

$f(x)$ is differentiable at $x = 0$

3(a) $\lim_{x \to 0} \frac{\tan x - x}{x} = \lim_{x \to 0} \frac{\sec^2 x - 1}{1}$ $\left[\frac{0}{0} \Rightarrow \text{L'Hopital Rule} \right]$

$= \lim_{x \to 0} (\sec^2 x - 1)$

$= 1 - 1$

$= 0$

3(b) let $f(x) = e^{2x}$

$f'(x) = 2e^{2x}$

given $x < y < \ln 2$, there $\exists c$ on (x, y) such that

$f'(c) = \frac{f(y) - f(x)}{y - x} = \frac{e^y - e^x}{y - x}$

$f'(\ln 2) = 2e^{2\ln 2} = 2e^{\ln 4} = 2(4) = 8$

since $f'(x)$ is an increasing function,

$f'(x) < f'(c) < f'(y) < f'(\ln 2)$

$\therefore f'(c) < 8$

$\therefore \frac{e^y - e^x}{y - x} < 8$

since $x < y$,

$y - x = -(x - y) = |x - y|$

$e^y > e^x$

$e^y - e^x > 0$

$e^y - e^x = |e^y - e^x|$

$\therefore \left| \frac{e^y - e^x}{x - y} \right| < 8$

4(a) $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$e^{x^3} = \sum_{n=0}^{\infty} \frac{(x^3)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^{3n}}{n!}$ } Taylor Series

let $f(x) = e^{x^3}$

$$e^{x^3} = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(60)}(0)}{60!}x^{60} + \dots$$

$$\Leftrightarrow \frac{f^{(60)}(0)}{60!}x^{60} = \frac{x^{3n}}{n!} \quad \text{where } 3n = 60$$

$$\Rightarrow n = 20$$

$$\therefore f^{(60)}(0) = \frac{60!}{20!}$$

(b) $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$

$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n$

$\frac{d}{dx}\frac{1}{1+x} = -\frac{1}{(1+x)^2}$

$$\frac{1}{(1+x)^2} = -\frac{d}{dx}\frac{1}{1+x}$$

$$= -\frac{d}{dx}\sum_{n=0}^{\infty}(-1)^n x^n$$

$$= \frac{d}{dx}\sum_{n=0}^{\infty}(-1)^{n+1} x^n$$

$$= \frac{d}{dx}\left[-1 + \sum_{n=1}^{\infty}(-1)^{n+1} x^n\right]$$

$$= \sum_{n=1}^{\infty}\frac{d}{dx}(-1)^{n+1} x^n$$

$$= \sum_{n=1}^{\infty}(-1)^{n+1} n x^{n-1}$$

$$= 1 - 2x + 3x^2 - 4x^3 + \dots$$

let $f(x) = 1 - 2x + 3x^2 - 4x^3$

$$\therefore \frac{1}{1.01^2} \simeq f(1.01 - 1)$$

$$= f(0.01)$$

$$= 0.9803$$

5(a) $f(x) = \dfrac{2+x^2}{x^2+2x+3}$

$$f'(x) = 2x(x^2+2x+3)^{-1} + (2+x^2)(-1)(x^2+2x+3)^{-2}(2x+2)$$
$$= \frac{2x(x^2+2x+3) - (2+x^2)(2x+2)}{(x^2+2x+3)^2}$$
$$= \frac{2x^2+2x-4}{(x^2+2x+3)^2}$$
$$= \frac{2(x+\frac{1}{2})^2 - \frac{9}{2}}{[(x+1)^2+2]^2}$$

let $f'(x) > 0$

$\Rightarrow 2(x+\frac{1}{2})^2 - \frac{9}{2} > 0$

$(x+\frac{1}{2})^2 > \frac{9}{4}$

$\Rightarrow x+\frac{1}{2} > \frac{3}{2}$ or $x+\frac{1}{2} < -\frac{3}{2}$

$x > 1$ or $x < -2$

interval where f is increasing $= \{x : x > 1 \text{ or } x < -2\}$

(b) $f(x) = \dfrac{2+x^2}{x^2+2x+3} = \dfrac{x^2+2x+3-2x-1}{x^2+2x+3} = 1 - \dfrac{2x+1}{(x+1)^2+2}$

since $x^2+2x+3 > 0$ for all $x \in \mathbb{R}$, there is no vertical asymptote

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty}\left[1 - \frac{2x+1}{x^2+2x+3}\right]$$
$$= 1 - \lim_{x\to\infty}\frac{2x+1}{x^2+2x+3}$$
$$= 1 - \lim_{x\to\infty}\frac{2}{2x+2}$$
$$= 1$$

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty}\left[1 - \frac{2x+1}{x^2+2x+3}\right]$$
$$= 1 - \lim_{x\to}\frac{2x+1}{x^2+2x+3}$$
$$= 1 - \lim_{x\to-\infty}\frac{2}{2x+2}$$
$$= 1$$

$\therefore$ $y=1$ is the horizontal asymptote

5(c) $f'(x) = \dfrac{2(x+\frac{1}{2})^2 - \frac{9}{2}}{[(x+1)^2+2]^2}$

let $f'(x) = 0$

$2(x+\frac{1}{2})^2 - \frac{9}{2} = 0$

$(x+\frac{1}{2})^2 = \frac{9}{4}$

$x+\frac{1}{2} = \pm\frac{3}{2}$

$x = -\frac{1}{2} \pm \frac{3}{2} = 1$ or -2

$f(1) = \dfrac{2+1}{1^2+2+3} = \dfrac{3}{6} = \dfrac{1}{2}$

$f(-2) = \dfrac{2+4}{4-4+3} = \dfrac{6}{3} = 2$

$\therefore$ The critical points are $(1, \frac{1}{2})$ and $(-2, 2)$

$f'(x) = \dfrac{2x^2+2x-4}{(x^2+2x+3)^2}$

$f''(x) = (4x+2)(x^2+2x+3)^{-2} + (2x^2+2x-4)(-2)(x^2+2x+3)^{-3}(2x+2)$

$= \dfrac{(4x+2)(x^2+2x+3) - 2(2x+2)(2x^2+2x-4)}{(x^2+2x+3)^3}$

$= \dfrac{-4x^3-6x^2+24x+22}{((x+1)^2+2)^3}$

let $g(x) = -4x^3-6x^2+24x+22$

$g(1) = 36 \Rightarrow f''(1) > 0$

$\therefore (1, \frac{1}{2})$ is a minima

$g(-2) = -16 \Rightarrow f''(-2) < 0$

$\therefore (-2, 2)$ is a maxima